Vector Addition

Exam code: 9MA0

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Help me understand

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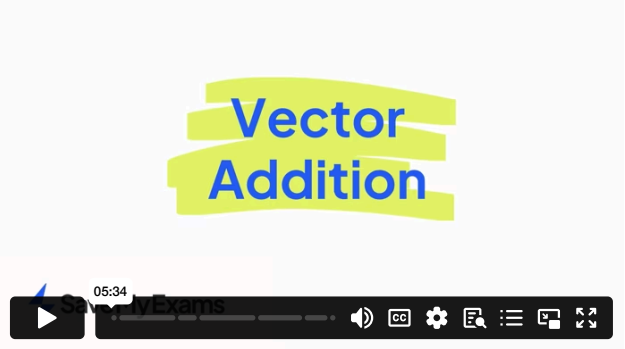
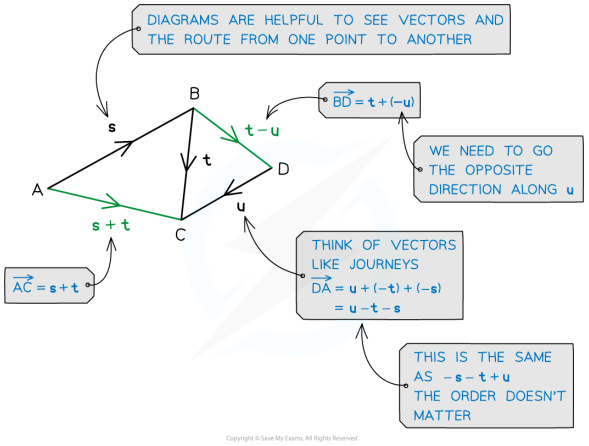


Did this video help you? Yes No

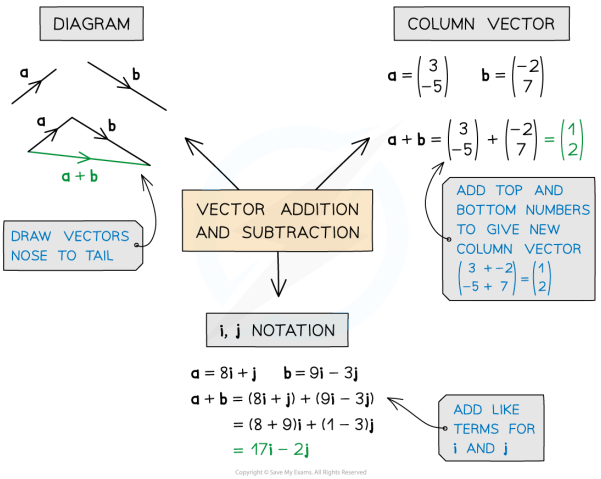
Vector addition

What is vector addition?

- **Adding** vectors together lets us describe the **movement between two points**
- The **single** vector this creates is called the **resultant** vector
- **Subtracting** a vector is the same as **adding the negative** vector
- Adding the vectors **PQ** and **QP** gives the zero vector, denoted by a bold zero **0**

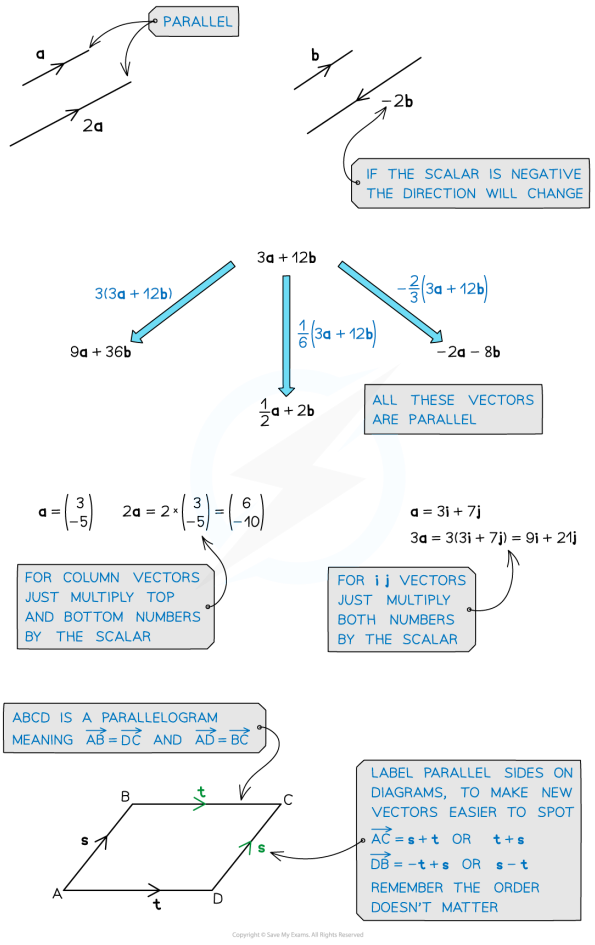


Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



Scalars and parallel vectors

- Multiplying by a positive **scalar** only changes the **size** of a vector, not its **direction**
- Two vectors are **parallel** if and only if one is a **scalar multiple** of the other



Examiner Tips and Tricks

- Think of vectors like a journey from one place to another – you may have to take a detour eg. A to B might be A to O then O to B.
- Diagrams can help, so if there isn't one, draw one. If there are any, labelling parallel vectors will help.

Worked Example

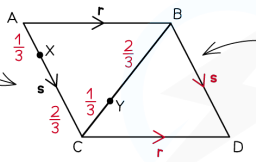
ABCD is a parallelogram with $\overrightarrow{AB} = \mathbf{r}$ and $\overrightarrow{AC} = \mathbf{s}$.

Point X lies on $\overrightarrow{AC}$ such that AX : XC = 1 : 2 and

point Y lies on $\overrightarrow{CB}$ such that CY : YB = 1 : 2

Show that $\overrightarrow{XY}$ is parallel to $\overrightarrow{AD}$.

AX:XC = 1:2 MEANS AX = $\frac{1}{3}$s AND XC = $\frac{2}{3}$s, LABEL THE DIAGRAM TO HELP



DRAW A DIAGRAM TO HELP, LABEL ALL PARALLEL SIDES

$\overrightarrow{AD} = \mathbf{s} + \mathbf{r}$

$\overrightarrow{CB} = \mathbf{r} - \mathbf{s}$

$\overrightarrow{XY} = \frac{2}{3}(\mathbf{s}) + \frac{1}{3}(\mathbf{r} - \mathbf{s})$

$\overrightarrow{XY} = \frac{2}{3}(\overrightarrow{AC}) + \frac{1}{3}(\overrightarrow{CB})$

THINK ABOUT THE SEPARATE PARTS OF THE "JOURNEY"

$\overrightarrow{XY} = \frac{2}{3}\mathbf{s} + \frac{1}{3}\mathbf{r} - \frac{1}{3}\mathbf{s} = \frac{1}{3}\mathbf{r} + \frac{1}{3}\mathbf{s} = \frac{1}{3}(\mathbf{r} + \mathbf{s})$

$\overrightarrow{XY} = \frac{1}{3}(\overrightarrow{AD})$

THEY ARE SCALAR MULTIPLES SO MUST BE PARALLEL

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